

Comptia A+ Assignment

Module -1, 2 [Hardware and its components]

RAM and Storage Theory.

**Types of RAMS (DDR3, DDR4, DDR5) and storage (HDD vs. SSD, NVMe).
Remove and reinstall RAM and storage drives.**

Task

❖ **1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.**

➤ Below given the differences between DDR3, DDR4, and DDR5 RAM.

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
Speed	800–2133 MHz	2133-3200 MHz	4800-8400+ MHz
Voltage	1.5V	1.2V	1.1V
Power Consumption	High	Lower than DDR3	Lowest and most power-efficient
Performance	Good for basic computer	Better for gaming, multitasking and productivity	Excellent for AI, 4K,8K editing, gaming

Maximum Memory Capacity	Lower	Higher than DDR3	Much higher than DDR4
Typical Use Cases	Older pc, office work, web browsing	Modern desktops, laptops, gaming, content creation	High-end gaming PCs, workstations, AI, video editing, and future-ready systems

❖ **2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully. Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.**

➤ Step 1: Shut Down the Computer

The computer was completely shut down and disconnected from the power supply. The battery (if removable) was also removed.



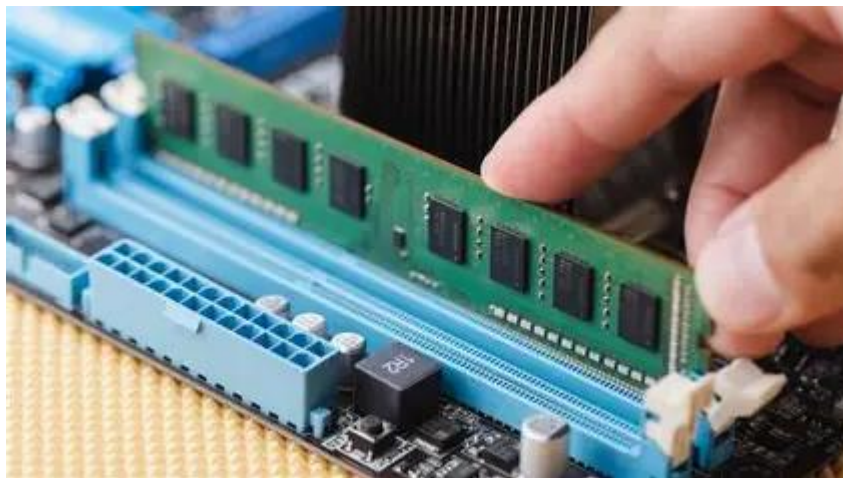
➤ Step 2: Open the System

The back cover (laptop) or side panel (desktop) was opened carefully using a screwdriver.



➤ Step 3: Locate the RAM Module

The RAM slot was identified on the motherboard.



➤ Step 4: Remove the RAM

The retaining clips on both sides of the RAM slot were gently pushed outward. The RAM module automatically lifted slightly and was removed carefully by holding its edges.



➤ Step 5: Reinstall the RAM

The notch on the RAM module was aligned with the slot, and the module was inserted firmly until both locking clips clicked into place.



➤ Step 6: Reassemble the Computer

The computer cover was closed securely, and all screws were tightened properly.

➤ Step 7: Verify System Boot

The power cable was connected, and the computer was switched on. The operating system loaded successfully without any hardware errors.



❖ **3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.**

➤ Below given the differences between HDD, SSD and NVMe storage.

Feature	HDD	SSD	NVMe
Speed	Slow (80-160 MB/s)	Fast (500-550 MB/s)	Very Fast (2000-7000+ MB/s)
Technology	Uses spinning magnetic disks	Uses flash memory (no moving parts)	Uses flash memory with PCIe interface for maximum speed
Cost & Best Use	Cheapest, large storage capacity	Moderate price with good performance	Most expensive but highest performance

➤ Below also given their Apps and use cases

Storage Type	Daily App / Use Case	Why It Benefits
HHD	Storing <u>Movies, Photos, and Backup Files</u>	<u>Large storage capacity</u> at a low cost.

SSD	<u>Google Chrome</u> or <u>Microsoft Office</u>	<u>Faster boot time and quicker application loading.</u>
NVMe	<u>Adobe Premiere Pro or Modern Games (e.g., GTA V, Valorant)</u>	<u>Extremely fast loading, file transfers, and high-performance tasks.</u>

❖ **4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly. Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.**

➤ Step 1: Power Off the Computer

The computer was shut down completely. The power cable and charger were disconnected before opening the system.



➤ Step 2: Open the Computer

The back cover (Laptop) or side panel (Desktop) was opened carefully using a screwdriver.



➤ Step 3: Locate the Storage Drive

The installed storage drive was identified. The connector type was observed as **SATA / M.2 / NVMe**



➤ Step 4: Remove the Storage Drive

The screws holding the storage drive were removed carefully. The drive was gently disconnected from the connector without applying excessive force.



➤ Step 5: Reinstall the Storage Drive

The storage drive was aligned correctly with the connector and reinstalled securely. The mounting screws were tightened properly.



➤ Step 6: Reassemble the Computer

The computer cover was closed and all screws were tightened securely.

➤ Step 7: Verify Storage Detection

The power cable was connected and the computer was switched on. The operating system started successfully, and the storage drive was detected correctly in the BIOS and Windows File Explorer.

